

Welcome To The Presentation

Submitted by

- H M Yeasin
ID: 20245103021
- Md. Fahim Islam
ID: 20245103035
- Md Sakibur Rahman
ID: 20245103091
- Abu Ahmad Mahfuzor Rahman
ID: 20245103092
- MD.Abdur Rahim
ID: 20234103011

Submitted to

Md. Ashik Khan

Lecturer

Department of Electrical and
Electronic Engineering

Topics Covered

- Introduction to Fire Detectors.
- Components Required for Fire Detectors.
- Circuit Diagram and Working Principle.
- Applications in Factory Security and Home Safety.
- Advantages and Limitations of Fire Detectors

Introduction to Fire Detectors And Purpose of it.

A **Fire Detection System** is an electronic setup designed to identify signs of fire such as smoke, heat, or flames and activate an alarm to alert occupants and trigger safety responses.

Purpose of Fire Detection System

- Detect fire hazards early to protect lives and reduce property damage
- Provide immediate and reliable alerts for fast evacuation and response
- Automatically activate safety systems such as sprinklers and alarms
- Continuously monitor environments to maintain ongoing protection
- Ensure compliance with safety regulations and improve risk management

Components :

BC547 Transistor (NPN) 1no

Infrared Receiver LED 1no

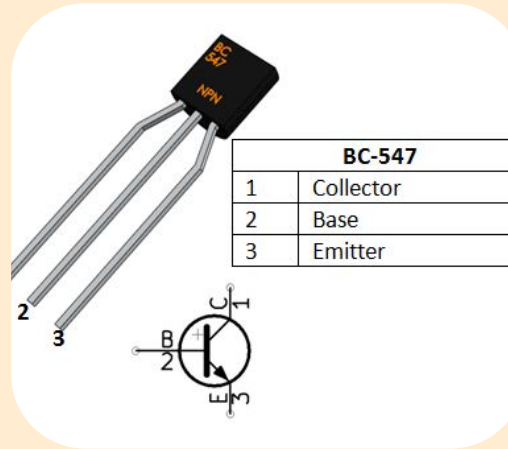
220-ohm Resistor 1no

LED 5mm 1no

DC Buzzer 1no

5 volt DC supply or 9 Volt Battery.

Breadboard

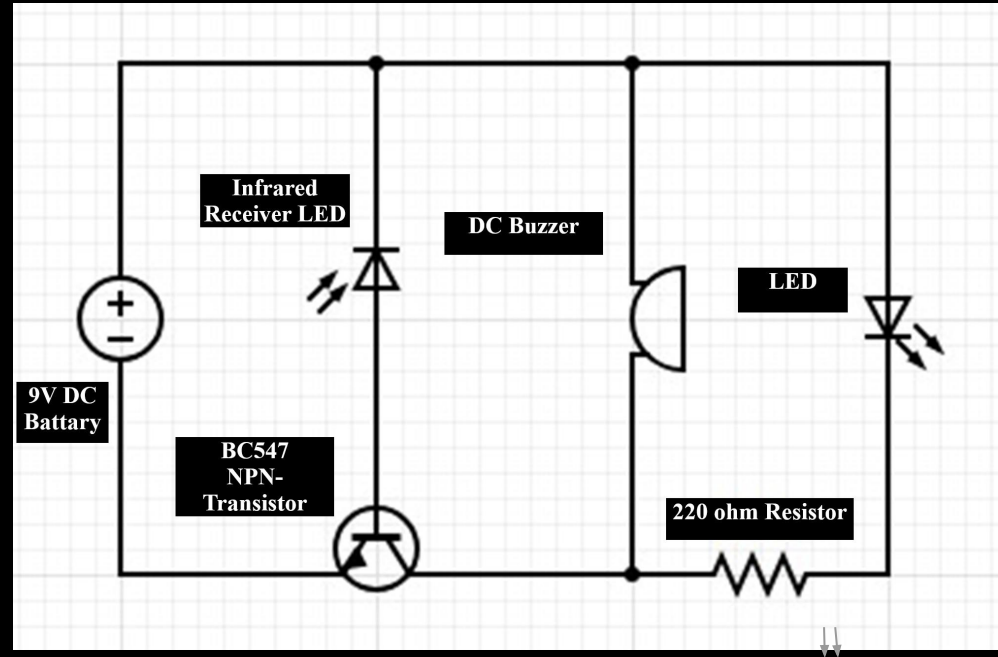


Circuit Diagram and Working

Principle:

1. **IR Receiver LED:** Detects infrared radiation (heat/flame) and sends a signal.
2. **BC547 Transistor:** Acts as a switch and amplifies the sensor signal.
3. **LED:** Lights up when fire is detected.
4. **Buzzer:** Sounds an alarm to alert people nearby.
5. **220Ω Resistor:** Limits current to protect components.
6. **Power Supply:** 5V or 9V battery powers the circuit.
7. **Breadboard:** Used to assemble and test the circuit without soldering.

Circuit Diagram :



An aerial photograph of a large solar farm with rows of solar panels stretching across the landscape. A black rectangular text box is positioned in the upper left quadrant, containing yellow and white text. The background image is a high-angle shot of solar panels, with some panels appearing to glow with a yellow-orange light, possibly due to reflection or a digital effect. The panels are arranged in neat, parallel rows that recede into the distance.

Applications of Fire Detection Systems in Factory Protection and Home Safety:

1. **Enable early detection** of fire hazards in both industrial and residential environments
2. **Monitor high-risk areas** such as manufacturing zones, kitchens, and electrical systems
3. **Provide immediate alerts** to support rapid evacuation and emergency response
4. **Integrate with safety systems** including sprinklers, alarms, and smart home devices
5. **Enhance occupant safety** by minimizing exposure to fire, smoke, and toxic gases
6. **Reduce potential losses** through fast containment and damage control
7. **Support compliance** with fire safety codes and regulations in factories and homes
8. **Deliver continuous protection** through 24/7 environmental monitoring

Limitations of Fire Detectors:

- **False Alarms (Light):** Flame or light-based sensors may trigger on reflections, sunlight, or welding arcs
- **Maintenance Needs:** Require regular testing, cleaning, and battery replacement
- **Very Low Detection Range:** Often limited to small areas; multiple units are needed for full coverage
- **Limited to Fire:** May only detect flame, smoke, or heat—cannot detect gas or chemical hazards
- **Environmental Interference:** Humidity, dust, airflow, or lighting conditions can reduce accuracy
- **Delayed Detection:** Time lag between ignition and sensor response, especially in large or ventilated areas
- **No Manual Stop:** Some systems rely only on automatic reset and lack a manual override
- **Limited to Alerts Only:** Typically provide sound and visual alarms; cannot suppress fire or auto-contact emergency services

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